

05 — Ablations and Causal Controls

SEION-KGR v26 MAX

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Purpose: report exactly which points of the preregistered 13-point causal ladder were executed, and what the executed subset can and cannot support.

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1 Preregistered ladder vs. executed

The full mandate causal ladder is 13 points (A0–A12) plus a 5-variant E8/control block (K0–K4), frozen verbatim in `preregistration.md` §4. This campaign executed exactly 2 of the 13 points, at screening scale:

Point	Definition	Executed this campaign?
A0	best_base	Yes — full dataset scale, 2 seeds, FB15K-237 + WN18RR
A1	+path_reasoner_standard_message	No — baseline variant not implemented
A2	+seionic_scalar	No
A3	+path_reasoner+seionic_message	Yes — 5,000-triple subsample, 2 seeds, both datasets
A4–A12	+projection/rank/selector/curricula/metaencoder	No — components exist and are unit-tested (Phases B2–B4, Fase 6–9) but not benchmarked at dataset scale in this campaign
K0–K4	E8/control block	No — implemented and unit-tested (Phase B3) but not run on benchmark data

2 Why only A0/A3

Per the mandate’s own Section X priority order, engineering closure (CI, selector, E8 branch, certification, evaluator audit) was prioritized above the full ablation ladder, and within Phase C, A0/A3 were chosen as the two most informative single comparison (does the combined path+seionic branch help at all, holding everything else fixed) given the compute available. Running the intermediate points (A1, A2 alone) or the projection/rank/curriculum/selector combinations (A4–A12) at dataset scale was not attempted — correctly reported here as **OPEN**, not silently absorbed into the A0/A3 comparison.

3 What the A0/A3 comparison can and cannot support

Can: demonstrate the architecture executes correctly end to end on real data with both branches active (already shown extensively in engineering smoke tests, Phases B2–B5); give a first, non-

confirmatory look at whether adding the path+seionic branches moves validation MRR in either direction on two real datasets.

Cannot: support any causal claim. Two seeds is explicitly insufficient for a variance estimate (reported per-seed only, §04); the A3 configuration ran on a 5,000-triple subsample of each dataset's training split (a compute-budget deviation logged *before* any A3 result was generated, `preregistration.md` §11), not the full training set A0 used — so any MRR difference between A0 and A3 is confounded with the training-data-size difference and cannot be attributed to the architectural change alone. This confound is stated explicitly here, not hidden in a footnote, and `04_gate12_confirmatory_benchmark_report.tex` repeats it next to the actual numbers.

4 Selector, geometric-loss, and E8 comparisons

Each of these has real, executed evidence at the *unit/engineering* level (learned selector: 14 tests including gold-path preservation after supervised warmup, `test_selector.py`; geometric-loss weight-gating: 4 regression tests proving zero compute at zero weight, `test_seion_kgr_package.py`; E8 vs. 4 matched controls: 19 tests including 2 against the real kernel, `test_structural_kernel.py`) but **none at dataset/ablation scale**. Reported as OPEN for the causal question ("does X measurably help on real KG link prediction"), NUMERICALLY_TESTED/DEFINITION for the engineering-correctness question ("does X work as specified").

5 Causal interpretation limits, stated once more

No result in this document, or this package, supports a statement of the form "SEION-KGR's {path reasoner / seionic branch / projection / E8 residual} causes better link prediction." That statement requires the full preregistered ladder at confirmatory scale (5 seeds, matched compute, statistical testing per `preregistration.md` §8) — not executed in this campaign.